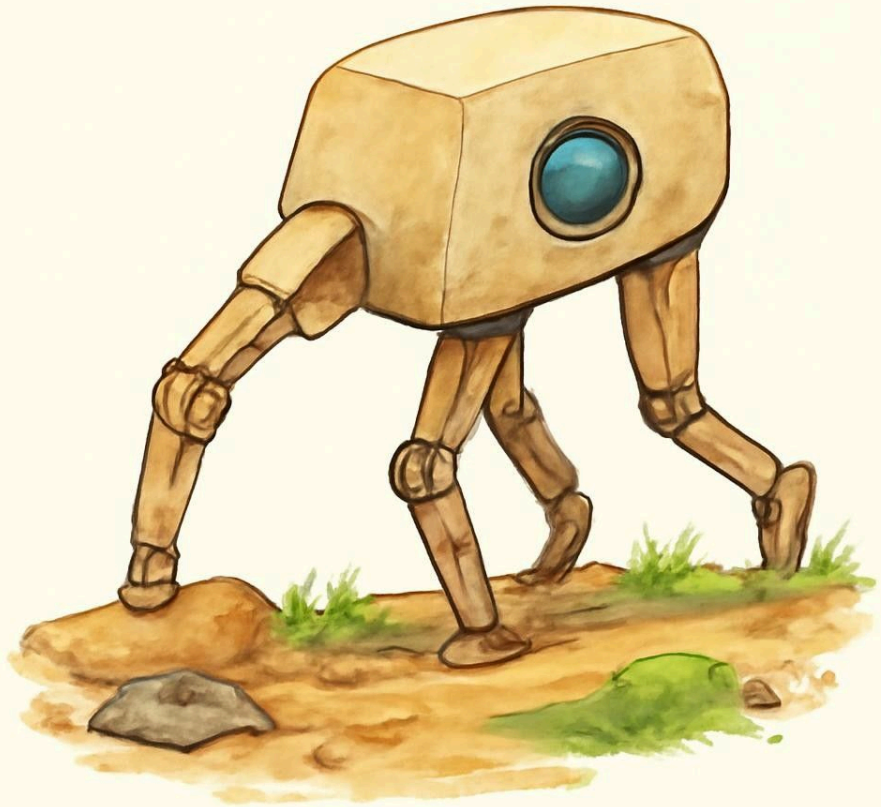


QUADRUPED ROBOT GAIT & BALANCE CONTROL

Course: BAEEE202 — Control Systems ·

Team 9 · Kartikay Singhal (25BEL0004)

Programme: B.Tech — Electrical & Computer Science ·

Semester: Fall 26–27

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What is a Quadruped Gait?

- **Quadruped robots:** Four legs for locomotion & stability
- **Robot gait:** Sequence and timing of leg movements
- **Gait planning:** Coordinates foot placement, joints & body motion
- **Stability:** Maintains balance during movement
- **Uneven terrain:** Requires adaptive and stable gait
- **Goal:** Achieve efficient locomotion with stability

Key chain: Gait Planning → Leg Coordination → Body Motion → Stable Locomotion

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Gait & Stability Challenges

- Uneven terrain and disturbances can cause **body instability**.
- Stability depends on **orientation, foot placement, gait phase, CoM, and ground contact**.
- **Pitch angle is selected** as the primary parameter as it directly indicates **forward–backward body balance**.
- This enables a **simple SISO control approach** for stability and disturbance recovery.

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Recent Literature Survey

- Bio-inspired DRL: gait switching & recovery across terrains (Humphreys & Zhou, 2025)
- Foot trajectory shapes gait patterns (Suzuki et al., 2025)
- Adaptive controllers for high-slope locomotion (Yin et al., 2024)
- Perturbation-adaptive learning (PA-LOCO, Xiao et al., 2024)
- Fall prediction & RL recovery strategies (Sun et al., 2024)

Research gap:

Existing studies demonstrate advanced gait adaptation and disturbance recovery, but there is limited systematic analysis of how fundamental gait parameters influence the trade-off between locomotion performance and stability under varying conditions.



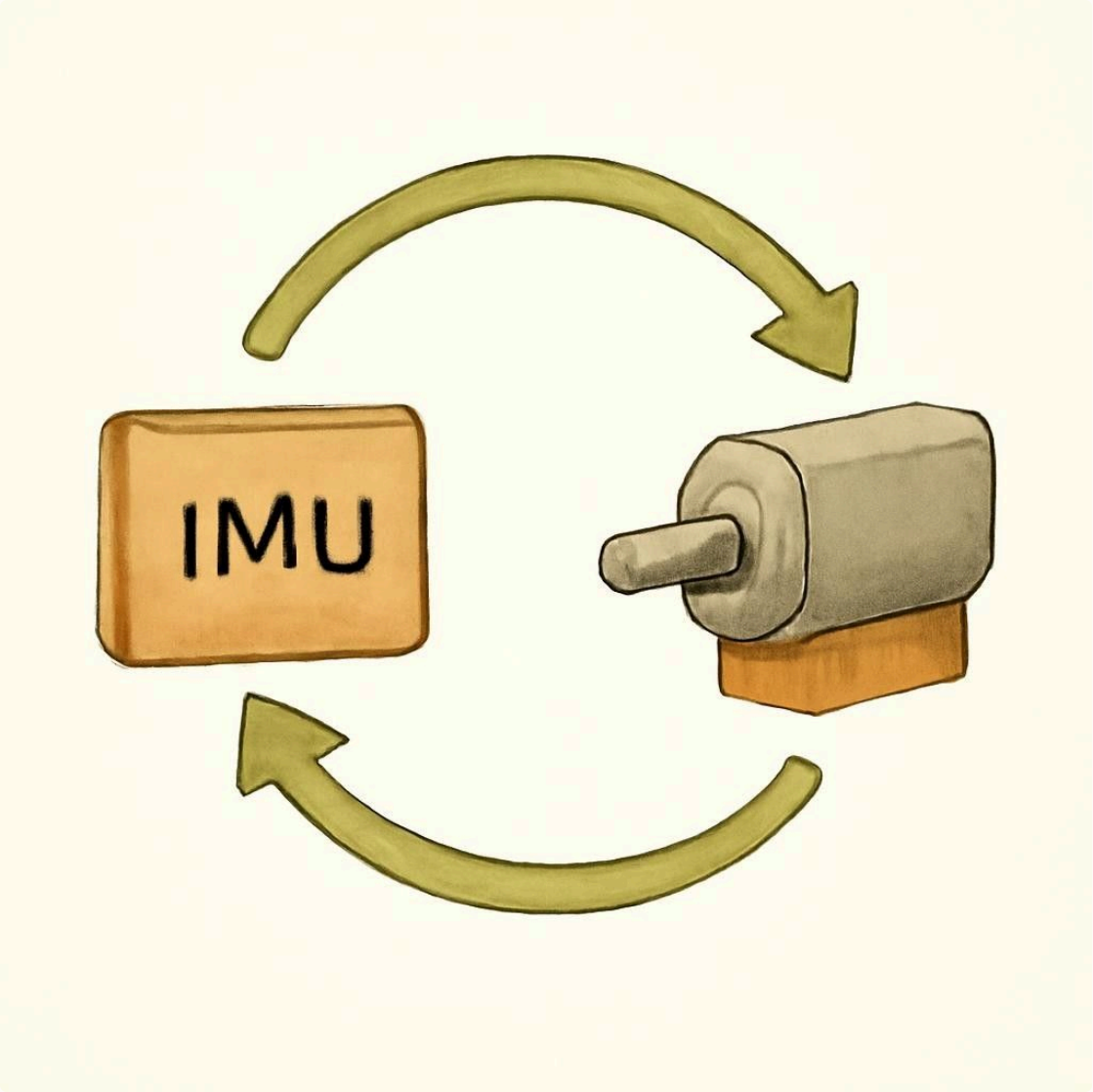
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Problem Statement

To develop and evaluate a coordinated gait strategy for quadruped locomotion using body pitch angle as the primary stability parameter, by analyzing and regulating the relationship between leg movement, foot placement, and body pitch to maintain a balanced body posture during forward motion



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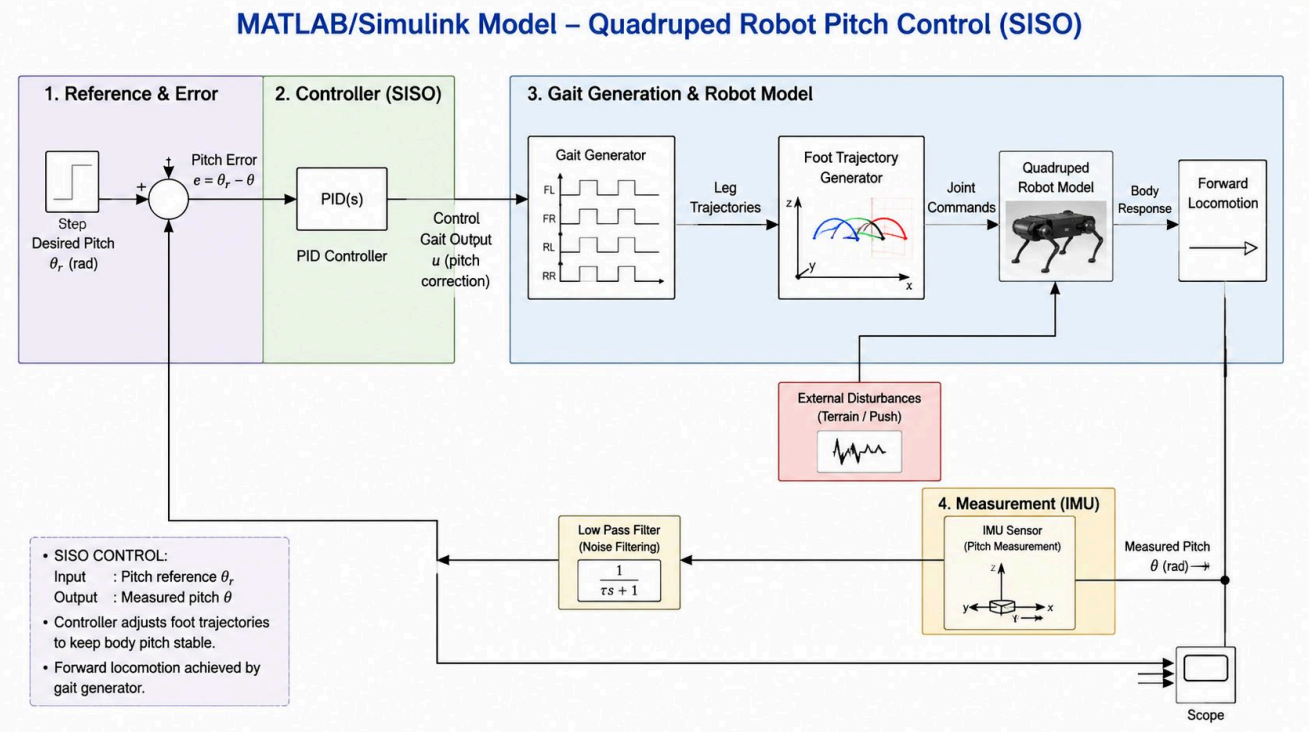
Project Objectives

- **Gait Development:**
Develop a coordinated gait for stable quadruped locomotion.

- **Pitch-Based Control:**
Use **body pitch angle** as the primary stability parameter.

- **Performance Evaluation:**
Test stability under **terrain variations and external disturbances**.

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Major Steps

1. Pitch Reference → Set desired pitch angle
2. Pitch Feedback → Measure actual pitch using IMU
3. SISO PID Control → Calculate pitch correction
4. Gait & Foot Trajectory → Generate coordinated leg motion
5. Robot Simulation → Execute gait and apply disturbances
6. Performance Evaluation → Analyze stability and locomotion

Overall flow:
Pitch Reference → Error → PID → Gait → Foot Trajectory → Robot → IMU → Filtering → Feedback → PID

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Two-Month Plan

Week 1 & 2

Lit. survey & requirement analysis — scope finalized

Variable & control selection — θ chosen, SISO loop defined

Week 3 & 4

Quadruped modelling in Simulink — joints & kinematics

Gait generation — leg sequence, timing, forward motion

Week 5 & 6

Foot trajectory implementation — convert to joint commands

SISO pitch controller integration — closed-loop tests

Week 7 & 8

Testing & tuning across terrains & disturbances

Validation, result comparison, documentation & PPT

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References & Resources

- **Journal / Conference Papers**

[1] J. Humphreys and C. Zhou, “Learning to adapt through bio-inspired gait strategies for versatile quadruped locomotion,” *Nature Machine Intelligence*, vol. 7, pp. 1141–1153, 2025, doi: 10.1038/s42256-025-01065-z.

[2] S. Suzuki, K. Matayoshi, M. Hayashibe, and D. Owaki, “Foot trajectory as a key factor for diverse gait patterns in quadruped robot locomotion,” *Scientific Reports*, vol. 15, Art. no. 1861, 2025, doi: 10.1038/s41598-024-84060-5.

[3] Z. Xiao *et al.*, “PA-LOCO: Learning Perturbation-Adaptive Locomotion for Quadruped Robots,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2024, doi: 10.1109/IROS58592.2024.10801753.

[4] H. Sun *et al.*, “Fall prediction, control, and recovery of quadruped robots,” *ISA Transactions*, vol. 151, pp. 86–102, 2024, doi: 10.1016/j.isatra.2024.05.039.

Software / Model Resources

[5] MathWorks, “Running Robot Model in Simscape,” *MATLAB Central File Exchange*. **MathWorks — Running Robot Model in Simscape**

[6] MathWorks, “Simscape-Robot-4Legs,” *GitHub repository*, 2023. **GitHub — MathWorks/Simscape-Robot-4Legs**

- **Textbooks**

[7] K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2010.

[8] R. C. Dorf and R. H. Bishop, *Modern Control Systems*, 13th ed. Pearson, 2016.

[9] N. S. Nise, *Control Systems Engineering*, 8th ed. Wiley, 2019.

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Q & A

Thank you — questions, suggestions, or clarifications?

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